HTO-1000
Thermal Oxidiser

User Manual
Version: 1.4
May 2007

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Manufacturers statement

Thank you for selecting the Ecotech HTO-1000 Thermal Oxidiser. The HTO-1000 is designed primarily for gaseous sampling using standard gas analysers. By following the guidelines contained in this manual and with the implementation of a good quality-assurance program, the user can obtain accurate and reliable data.

This User Manual provides a complete product description including operating instructions, calibration, and maintenance requirements for particulate sampling techniques.

If, after reading this manual you have any questions or you are still unsure or unclear on any part of the HTO-1000 then please do not hesitate to contact Ecotech.

Ecotech also welcomes any improvements that you feel would make this a more useable and helpful product then please send your suggestions to us here at Ecotech.

Notice

The information contained in this manual is subject to change without notice. Ecotech reserves the right to make changes to equipment construction, design, specifications and /or procedures without notice.

WARNING

Hazardous voltages exist behind the sampler control panel and within the sampler housing. Ensure the control panel and front and rear enclosure panels are always in place when the sampler is connected to the mains. The front doors should be closed when the sampler is left unattended. Ensure the power cord, plugs and sockets are maintained in a safe working condition.

Protective earth straps fitted to doors and within the electronics enclosure must be connected at all times when power is connected to the sampler.

Ecotech recommends the use of Earth-Leakage Protection Circuit Breakers (ELCB) on the power supply to the GasCal-1100. If operating from a generator, power conditioner, isolation transformer, or other floating supply, refer to the section on Instrument setup, as some important modifications are required.

Safety requirements

- To reduce the risk of personal injury caused by electrical shock, follow all safety notices and warnings in this documentation.

- If the equipment is used for purposes not specified by Ecotech, the protection provided by this equipment may be impaired.

- Replacement of any part should only be carried out by qualified personnel, only using parts specified by Ecotech. Always disconnect power source before removing or replacing any components.

Factory service/warranty

This product has been manufactured with care and attention.

The product is subject to a 12-month warranty on parts and labour. The warranty period commences when the product is shipped from the factory. Consumable items are not covered by this warranty.

To ensure that we process your factory repairs and returned goods efficiently and expeditiously, we need your help. Before you ship any equipment to our factory, please call your local Ecotech service response centre (or distributor) to obtain a return authorisation number.

When you call please be prepared to provide the following information:

1. your name, telephone number and Facsimile number
2. Your company name
3. The model number or a description of each item
4. The serial number of each item, if applicable
5. A description of the problem or the reason you are returning the equipment (eg, sales return, warranty return, etc)

If you are required to return the equipment an accompanying document with:

1. Your name, number and Facsimile number
2. Your company name with return shipment
3. The model number or a description of each item
4. The serial number of each item, if applicable

A description of the problem/reason you are returning the equipment

Claims for Damaged Shipments and Shipping Discrepancies

Damaged shipments

1. Inspect all instruments thoroughly on receipt. Check materials in the container(s) against the enclosed packing list. If the contents are damaged and/or the instrument fails to operate properly, notify the carrier and Ecotech immediately.

2. The following documents are necessary to support claims:

   a. Original freight bill and bill lading
   b. Original invoice or photocopy of original invoice
   c. Copy of packing list
   d. Photographs of damaged equipment and container

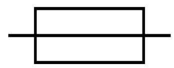
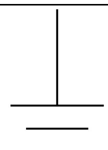
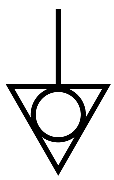
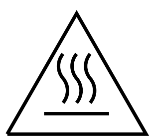
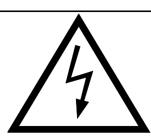
   You may want to keep a copy of these documents for your records also.

   Refer to the instrument name, model number, serial number, sales order number, and your purchase order number on all claims. Upon receipt of a claim, we will advice you of the disposition of your equipment for repair or replacement.

Shipping Discrepancies

Check all containers against the packing list immediately on receipt. If a shortage or other discrepancy is found, notify the carrier and Ecotech immediately. We will not be responsible for shortages against the packing list unless they are reported promptly.

Internationally recognised symbols used on Ecotech Equipment

	Electrical fuse	IEC 60417, No. 5016
	Earth (ground) terminal	IEC 417, No. 5017
	Equipotentiality	IEC 417, No. 5021
	Alternating current	IEC 417, No. 5032
	Caution, hot surface	IEC 417, No. 5041
	Caution, refer to accompanying documents	ISO 3864, No. B.3.1
	Caution, risk of electric shock	ISO 3864, No. B.3.6

Manual Revision History

Ecotech Manual ID: **MAN 0037**
Manual PN: **ECO-HTO1000MAN**
Current Revision: **1.4**
Date released: **May 2007**
Description: **User Manual for the HTO-1000 Thermal Oxidiser**

This manual is a newly created document for use of the HTO-1000 Thermal Oxidiser.

Edition	**Date**	**Summary**	**Affected Pages**
1.0	January 2003	Release	
1.1	July 2003	Minor corrections	various
1.2	January 2004	HC configuration added	various
1.3	February 2006	Temperature ranges updated	2, 11
1.4	May 2007	Manual redesigned	All

Table of Contents

1 Introduction

1.1 Description

The Ecotech HTO-1000 Thermal Oxidiser is a general-purpose thermal converter designed to be used with chemiluminescent oxides of nitrogen gas analysers for the measurement of NH_3 Ammonia and NO2 nitric dioxide, or with sulfur dioxide gas analysers for the measurement of H_2S hydrogen sulfide or TRS (Total Reduced Sulfur.) TRS comprises hydrogen sulfide (H_2S), methyl mercaptan (MESH), dimethyl sulfide (DMS) ethyl mercaptan and dimethyl disulfide (DMDS). The Ecotech Model HTO-1000 series of thermal oxidisers is specifically configured for each gas application with different catalytic materials and operating temperatures. The converter is available with a SO_X scrubber where thermal oxidation of chemical compounds requires temperatures in excess of those unobtainable with low temperature ovens. The oxidiser is generally used to oxidise gaseous compounds in an atmosphere containing oxygen.

The HTO-1000HC is designed to remove light hydrocarbons such as methane, ethane, propane, etc.

1.2 Specifications

1.2.1 Converter Efficiency

Range:

- Ammonia: 88-94%
- Hydrogen Sulfide: 85-95%
- Total Reduced Sulfur: 95-99%
- Hydrocarbon Removal: <20HC per 10 litre/minute

Response time:

- 5-10% of rise/fall time of analyser.

Operating temperature

- 25-1000°C

1.2.2 Flow

Flowrate:

- 0.1 – 1.5L/min

Gas Temperature Accuracy:

- ±5 °C

Gas Pressure:

- From vacuum to max 2 bar pressure

Quartz tube volume:

- 88.5 cc

1.2.3 Power

Operating voltage:

- 240VAC, 50Hz fuse at 3.15A Slo-Blow
- 110 VAC, 60Hz fuse at 5A Slo-Blow

Consumption

- 200VA

1.2.4 Physical dimensions

Case dimensions:

- LxWxH = 220 x 484 x 220 mm (5RU)

Weight:

- 10kg

2 Installation

This section is intended to instruct the user in the proper installation and set-up procedures for the installation and commissioning of the converter.
The thermal oxidiser is housed in a rack-mountable case that can be fitted with feet to make a bench-top unit. Since it operates at elevated temperatures, proper ventilation is required. For bench mounted installations, be sure never to place anything on top of the oxidiser. For panel rack mounting installations be sure to allow the minimum of at least 7.5-cm between top of oxidiser and bottom of instrument above it. In rack type installations it is preferred that the oxidiser be the instrument mounted at the top. For ventilation calculations use heat dissipation factor of 400 watts. In general the unit should be located in an environment free of dust, moisture, and vibration. Sample tubing should be kept as short as possible and only Stainless Steel glass line tubing or thick walled Teflon tubing should be used, in order to keep absorption losses to a minimum.

Warning: Never place objects on top of HTO-1000 and ensure proper ventilation of unit is always maintained.

2.1 Initial check

Check the unit for shipping damage. If damages has been found follow standard procedure and inform shipping company of damages that has occurred. Remove top covers and check the oven area of converter.

Note: The quartz conversion tube must be inspected to verify its integrity.

Operation of the instrument is relatively simple. To verify normal operation prior to installation, perform the following checks.

1. Verify that correct operating power has been selected for your application. If the instrument is operated at 220-240 50 Hz VAC you require to install a 2.5 amp sloblow fuse, if it is operated at 115 VAC 60 Hz you require a 5 amp sloblow fuse. Verify that unit is configured for the same voltage as the power source that is to be connected to.
2. Inspect quartz tube (non-HC models only) and ensure that it is intact. Replace top cover.
3. Plug power cord into power receptacle and into power source; turn on power at rear of unit. The temperature indicator on the front panel will now display current ambient temperature.
4. Set temperature controller to 200°C using the up keys on the temperature controller. The red indicator for “ON” should light on the temperature controller front panel.
5. After ten minutes the temperature should reach 200°C and control of heating should turn “ON/OFF”.
6. After the oven has stabilized adjust the temperature to 400°C and allow oven to stabilize at this new temperature setting.

7. Now set the oven at desired operating temperature and allow it to stabilize. If temperature does not stabilize follow tuning instructions of temperature controller for this new temperature setting.
8. Allow clean air to flow through the converter for at least one hour at the operating temperature before using instrument. This will allow any residual compound to be converted before usage. Zero air should be measured on the analyser ±0.020ppm max. The analyser should then null this zero out.
9. Following the zeroing of the analyser a span gas in the range of 80% of full scale should be used to compensate for any losses due to the installation of converter.

2.2 Assembly: Pneumatic Connections

The sample gas plumbing connections are easy to make. Two ¼" O.D. tube fittings are located at rear panel of converter. One is labelled "SAMPLE IN" this connection is for the sample gas to be converted; the other is labelled "SAMPLE OUT" for the oxidised gas to be routed to the gas analyser of choice. For H_2S and measurement connect the sample gas direct to SO_X scrubber and connect the output of SO_X scrubber direct to the sample-inlet of the oxidiser.

Note: It is very important to leak-check the pneumatic connections of converter and analyser before proceeding with measurement. Connect analyser exhaust to an appropriate exhaust system.

For NH_3, NO_2 and TRS connect the sample gas direct to sample-inlet of oxidiser. For NH_3 measurement, install the HTO-1000N on the N_X channel. Use a proprietary NH_3 scrubber when zeroing the instrument. In stack applications use an ammonia scrubber on the N_X inlet between the HTO-1000N and the N_X port and also at the NO_X channel inlet.

For HC removal, connect the incoming air to the SAMPLE IN port. Hydrocarbon free air will exit the SAMPLE OUT port. The outlet has an extension tube that allows the gas time to cool down. Do not reverse the connections or the exhaust gas will be too hot

3 Operation

The HTO-1000 is factory configured for a particular type of gas conversion. Details of the different applications are provided below.

3.1 Ammonia Measurement

The model HTO-1000N can be factory configured for the measurement of ammonia (NH_3). The oven temperature required for the conversion of NH_3 to NO should be set in the range of 580°C to 820°C catalyst dependant.
In this temperature range the conversion of NH_3 to NO is the most efficient, however some NH_3 will be converted to NO_2. The conversion process occurs when the gas sample flows through the quartz tube and the catalyst material set to 720°C, all of the oxides of nitrogen (N_X) are thermally oxidised to nitric oxides and some to nitrogen dioxide. The hydrogen molecules are additionally stripped from the NH_3 gas sample and converted H_2O and the nitrogen molecules recombine with oxygen to form NO and NO_2.
Converting all oxides of nitrogen (NO_X) and ammonia (NH_3) to NO using the converter then measure ammonia as NO. The instrument used to monitor ammonia is the Ecotech model EC9842 chemiluminescent NH_3 analyser. The model HTO-1000 N converter is plumbed to the N_X input, the difference between the NO_X = (NO + NO_2) and the N_X = (NH_3 + NO_2 +NO) channel is the NH_3 each channel has a separate gain in order to compensate for losses in the sampling loops.
The average thermal converter efficiency for ammonia conversion is 88-94%.

3.2 Hydrogen Sulfide and Total Reduced Sulfur Measurement

The model HTO-1000 is factory configured for the measurement of Hydrogen Sulfide (H_2S) or Total reduced Sulfur (TRS measurement). For the hydrogen sulfide H_2S measurement the oxidiser oven assembly requires a SO_X scrubber to be used with the converter.
H_2S is measured by converting H_2S with the thermal oxidiser to sulfur dioxide (SO_2) and measuring the resultant gas with an Ecotech EC9850 SO_2 analyser. The sample gas is passed through the SO_X scrubber on the inlet of the HTO-1000. The SO_X scrubber is installed to permit scrubbing of oxidised sulfur compounds in the sample stream. The SO_X and the catalyst will yeald a > 97% removal of SO_X compounds. Any losses in the converted gas <1% will be nulled during the calibration procedure of the EC9850 or analytical instrument of choice. The sample gas after passing through the SO_X scrubber where the sample gas is now only H_2S. The gas is then passed through the quartz tube, which is heated, to >560°C, stripping the hydrogen from the H_2S in the sample and reducing it to SO_2 and H_2O.
The oven temperature required for the conversion of H_2S to SO_2 should be set in the range of 560°C to 620°C for an average thermal reduction efficiency of 95 – 98 %.
For TRS Measurement there is no requirement for the use of the SO_X scrubber the design of the quartz converter complies with U.S. EPA method for the Determination of Total Reduced Sulfur found in the Federal Register 40 CFR Ch1, Pt60, App A, Method 16A (7-1-85 edition). The use of a quartz tube is derived from this method, the operating temperature recommended by U.S.EPA guidelines mentioned in Section 3.4, the operating temperature for a Total Reduced Sulfur conversion temperature is

800°C ±100°C. For gasses other than sulfur compounds, conversion temperature are to be determined by the end user. As this is a general-purpose device Ecotech Pty Ltd cannot be held responsible for its misuse or misapplication. The sample gas must contain at least 1-% oxygen for complete oxidation of all TRS to sulfur dioxide (SO_2). The lower detectable limit is 0.002ppm when the sample flow is 0.750 l/min and the upper concentration limit is dependent on the resident time of sample gas in the oxidiser.

3.3 Hydrocarbon removal

The model HTO-1000HC has been factory configured for the removal of light hydrocarbons from air. It is commonly used to generate hydrocarbon free zero air. The oven temperature required for the oxidation of light hydrocarbons should be set to 520°C.

Typical performance is <20ppb HC at up to 10 litres per minute of ambient air, however performance will vary depending on the original concentration and the flow rate.

Hydrocarbons are catalytically oxidised, forming mainly CO_2 and H_2O. The remainder of the zero air system can remove these impurities if required.

4 Calibration

4.1 TRS Calibration

Connect suitable Dilution Gas Calibrator such as the Ecotech model GasCal-1100 to the input of thermal converter. Allow 15 minutes for analyser to stabilize on zero gas.

Note: when the thermal converter is initially turned on, gasses and material imbedded on the quartz tube will usually take 60 minutes to burn off. As material burns off it may appear as SO_2 or NO.

Perform analyser zero calibration, the resulting zero offset generated by the converter should not exceed 0.100ppm. Introduce span calibration gas 80% of working range; perform a span calibration on the analyser. Check zero again and adjust if necessary.
Optimizing thermal converter temperature, introduce span gas of 100% of operating range. Allow thermal converter to stabilize, then increase thermal converter temperature by 25°C allow gas analyser value to stabilize (15-25 minutes). As the temperature of the converter is increased the span output will also increase until the "optimum thermal converter temperature" is reached; further increase in temperature will not increase conversion efficiency and will start to decrease. For long life of heaters and catalyst material it is better to run the oxidation temperature at an efficient level but not too high. If the temperature is too high efficiency will not be increased and the life of the catalytic material will be limited. The life of catalytic material and quartz tube should last several years under normal usage. Multi point calibration of the thermal converter is recommended at start-up and once per year.

4.2 NH_3 Calibration

Connect Dilution Gas Calibrator such as the Ecotech model GasCal-1100 to the input of thermal converter with FEP thick walled new tubing, and use 1/8 glass lined SS tubing from the NH_3 cylinder to the span input of GasCal-1100, keep all tubing as short as possible.

Important: Use only new Teflon tubing.

Allow 30 minutes for analyser to stabilize on zero gas. Note when the thermal converter is initially turned on, gasses and material imbedded on the quartz tube and catalyst will usually take 60 minutes to burn off. As material burns off it may appear as NO_X or NO. Perform analyser zero calibration, the resulting zero offset generated by the converter should not exceed 0.100ppm. Introduce span calibration gas 80% of working range; perform a span calibration on the analyser.

Note: NH_3 span gas will be slow to respond

Use only new glass lined SS tubing from the cylinder to GasCal-1100 and a high purity Stainless Steel regulator. Condition the tubing by running calibration gas through the converter for a period of several hours.
Response time of analyser may be check with NO gas. Check zeros again and adjust if necessary. Optimizing thermal converter temperature, introduce span gas of 100% of operating range. Allow thermal converter to stabilize, then increase thermal converter

temperature by 25°C allow gas analyser value to stabilize (30-60 minutes). As the temperature of the converter is increased the span output will also increase until the "Optimum thermal converter temperature" is reached; further increase in temperature will not increase conversion efficiency and will start to decrease. Typical NH_3 converter temperature is between 580-820°C (catalyst dependant). For long life of heaters and catalyst material, it is better to run the oxidation temperature at an efficient level, but not too high. If the temperature is too high efficiency will not be increased and the life of the catalytic material will be limited. The life of catalytic material and quartz tube should last several years under normal usage. Multi point calibration of the thermal converter is recommended on start-up and once per year.

5 Maintenance

5.1 Maintenance tools

To perform general maintenance on the HTO-1000 the user will be required to carry the following equipment:

- Distilled water
- Decom 90 lab detergent (Cleaning solution diluted 20:1 with distilled water)
- Glass beaker
- Liquid trap
- Teflon tubing
- Pump

5.2 Maintenance schedule

Interval (samples)*	Item	Procedure	Page
As needed	SO_X Scrubber Efficiency	Check	11
As needed	Thermal Converter Oxidising Tube	Clean	12

* Suggested intervals for maintenance procedure may vary with sampling intensity and environmental conditions.

5.3 Maintenance procedures

5.3.1 SO_X Scrubber Efficiency

The purpose of the SO_X scrubber is designed to remove all oxides of Sulfur found at the average sample concentrations for a period of six months or a total of 300-ppm hours. If the SO_X scrubber is suspect a spare unit should replace it or follow the following scrubber test procedure:

1. Zero the EC9850 analyser after a 15-minute of zero airflow.
2. Supply a known concentration of H_2S typically 80% of measuring range, and calibrate the EC9850 to read the exact same concentration.
3. Replace H_2S source with SO_2 and span with approximately 250 – 400 ppb SO_2.

4. Verify that the analyser does not show more than 5-ppb response to the SO_2 input if it's higher than the scrubber must be replaced.

5.3.2 Temperature Controller Set-up

The temperature controller in the Ecotech model HTO-1000 Thermal oxidiser has been factory set and tuned to operate as required. With the exception of operating temperature, no other parameter is required to be altered. For temperature controller set-up and tuning instructions refer to controller manual.
Set point is shown on the bottom line and the actual furnace temperature is shown on the top line.

5.3.3 Cleaning the Thermal Converter Oxidising Tube (not HC version)

If the conversion efficiency drops to below 80% on the thermal oxidiser, or the oxidiser has operated continuously for 12 months, cleaning the sample lines of the converter and the thermal oxidising tube may correct the problem. Follow the procedure below:

1. Turn off power to the model HTO-1000. Disconnect power cord from instrument. Disconnect sample-input port from source, disconnect sample outlet port from analyser, and also disconnect tubing to scrubber if scrubber is fitted and remove scrubber.
2. Allow thermal converter to cool for at least 4 hours. Do not proceed until thermal oxidiser cover and tubes are cool to the touch.
3. Connect sample inlet port of oxidiser to the liquid trap; connect the vacuum port of liquid trap to the pump.
4. Connect Teflon tubing to Sample outlet port of oxidiser. Place the other end of tubing in the beaker with the 20:1 solution.
5. Start pump. Allow pump to run until there is no solution left in the beaker or until liquid trap becomes full. Repeat this process until tubing and quartz tubing are cleaned.
6. Refill beaker with distilled water and repeats process at least three times, then allow pump to run at least 5 minutes to remove any moisture left in the assembly.
7. Connect zero air source to thermal converter (1/min) connect power cord to the model HTO-1000 and allow oxidiser to stabilize at operating temperature for at least 2 hours.
8. Remove zero air source, reconnect sample inlet to measurement source, reconnect sample outlet to analyser. (If option is fitted reconnect SO_X scrubber to the inlet of oxidiser, it is normal for SO_X scrubber to be replaced when cleaning is performed.
9. Allow the model HTO-1000 to stabilize flow and temperature for two hours.
10. Perform a zero/span calibration following the calibration a converter efficiency test is performed.